

AN ASYMPTOTIC DISTRIBUTION FOR THE COEFFICIENT IN A MULTIPLE TIME SERIES

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SUMMARY

The limiting distribution of the ML estimator $\hat{\beta}$ of β in the first-order stochastic equation $Y_t + \beta Y_{t-1} = U_t$ was derived by Anderson, T.W. (1971): The Statistical Analysis of Time Series. In this paper an asymptotic distribution for $\hat{\beta}$ to $O(N^{-3/2})$ will be derived which is more appropriate in the case of small samples.

1. Introduction

The purpose of this study is to find an asymptotic expression for the distribution of the estimator of the coefficient in a first-order p -component vector stochastic difference equation. The model and limiting distribution are given by Anderson (1959), (1971) and Anderson and Rubin (1950).

Theorem 1.1 (Anderson 1971), p. 188-200)

If $\{Y_t\}$ ($p \times 1$) is a sequence of random vectors satisfying $Y_t + \beta Y_{t-1} = U_t$ with $\{U_t\}$ independently and identically distributed with $E(U_t) = 0$ and $E(U_t U_t') = \Sigma$, if $-\beta$ ($p \times p$) has all characteristic roots less than one in absolute value, and if F is positive definite, then $\sqrt{N}(\hat{\beta} - \beta)$ has a limiting normal distribution with mean 0 and covariance matrix $\Sigma \otimes F^{-1}$, with

$$(1.1) \quad \hat{\beta} = - \left(\sum_{t=1}^N Y_t Y_{t-1}' \right) \left(\sum_{t=1}^N Y_{t-1} Y_{t-1}' \right)^{-1}$$

$$(1.2) \quad F = E(Y_t Y_t')$$

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and

$$(1.3) \quad F - \beta F \beta' = \Sigma.$$

2. An Asymptotic Characteristic Function

Consider the process under the assumption of normality and $Y_0 = 0$. Let Y denote the $(N+1) \times 1$ vector obtained when the process has been observed for $N+1$ periods and the observations placed in one column vector. Then

$$(2.1) \quad Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_{N+1} \end{bmatrix} \sim N(0; \Xi)$$

where $\Xi((N+1) \times (N+1))$ is given by

$$(2.2) \quad \Xi = \begin{bmatrix} F & -\beta F & \dots (-\beta)^N F \\ -F\beta' & F & \dots (-\beta)^{N-1} F \\ \vdots & \vdots & \ddots \\ F(-\beta')^N & F(-\beta')^{N-1} & \dots F \end{bmatrix}$$

Let

$$(2.3) \quad X_t(2 \times 1) = \begin{bmatrix} Y_{t+1} \\ Y_t \end{bmatrix}, \quad t = 1, 2, \dots, N.$$

and

$$(2.4) \quad X(2 \times N) = (X_1, \dots, X_N) = \begin{bmatrix} X^{(1)} \\ X^{(2)} \end{bmatrix} \quad \begin{matrix} (p \times N) \\ (p \times N) \end{matrix}$$

so that $\hat{\beta}$ defined in (1.1) can be written as

$$(2.5) \quad \hat{\beta} = -X^{(1)'} X^{(2)'} (X^{(2)} X^{(2)'})^{-1}$$

Also

$$(2.6) \quad E(XX') = N \begin{bmatrix} F & -\beta F' \\ -F\beta' & F \end{bmatrix} = N\Sigma_0, \text{ say.}$$

To examine the distribution of X , let x_c be the $2pN \times 1$ column vector formed when all the columns of X are packed in one column below each other.

Then

$$(2.7) \quad x_c = MY$$

where M is a $2pN \times p(N+1)$ matrix of $2N \times (N+1)$ submatrices of order $p \times p$. The i, j -th submatrix of M, M_{ij} , is equal to I_p when

$$(i) \quad i \text{ is even and } j = \frac{i}{2}, \text{ or}$$

$$(ii) \quad i \text{ is odd and } j = \frac{i+3}{2}$$

The rest of the submatrices are equal to $O(p \times p)$.

Thus x_c has a singular normal distribution (Anderson (1958)) with $E(x_c) = O$ and

$$(2.8) \quad E(x_c x_c') = M \Xi M' = \Xi^*$$

where Ξ^* ($2pN \times 2pN$) is singular.

Then

$$(2.9) \quad \phi_{\frac{XX'}{\sqrt{N}}}(T) = |I_{p(N+1)} - 2i \frac{M' \tau M \Xi}{\sqrt{N}}|^{-\frac{1}{2}} \text{ where } \tau = I_N \otimes T.$$

Consider now the matrix $\Xi^* = M \Xi M'$ which can be written in the form

$$(2.10) \quad \Xi^* = \begin{bmatrix} \Sigma_0 & \Sigma_1 & \Sigma_2 & \dots & \Sigma_{N-1} \\ \Sigma_1' & \Sigma_0 & \Sigma_1 & \dots & \Sigma_{N-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma_{N-1}' & \Sigma_{N-2}' & \Sigma_{N-3}' & \dots & \Sigma_0 \end{bmatrix}$$

where Σ_0 is given in (2.6) and

$$(2.11) \quad \Sigma_i = \begin{bmatrix} F(-\beta')^i & F(-\beta')^{i-1} \\ F(-\beta')^{i+1} & F(-\beta')^i \end{bmatrix}, \quad i = 1, 2, \dots, N-1$$

Let the covariance between Y_t and Y_{t-i} tends to zero for $i > k$, that is, $(-\beta)^k \rightarrow 0$. This means that the observation at time t can be treated as independent of observations before time $t-k$. k is a positive integer smaller than N . When $k \geq N$, k is replaced by N . This assumption is valid under the condition (Anderson (1971)) that the characteristic roots of $-\beta$ are smaller than one in absolute value.

Then

$$(2.12) \quad \Sigma_i = 0 \quad \text{for } i > k$$

Let

$$(2.13) \quad U = \sqrt{N} \left(\frac{XX'}{N} - \Sigma_0 \right),$$

then

$$(2.14) \quad \phi_U(T) = \text{etr}(-i\sqrt{N}\Sigma_0 T) \text{etr}\left(\frac{i}{\sqrt{N}} \text{tr} M' \tau M \Xi\right) \text{etr}\left(-\frac{1}{N} (M' \tau M \Xi)^2\right) \\ \cdot \left\{ 1 - \frac{i}{6\sqrt{N}N} \text{tr} (M' \tau M \Xi)^3 + O(N^{-2}) \right\}$$

Now

$$(2.15) \quad \text{tr} M' \tau M \Xi = N \text{tr} \Sigma_0 T$$

and

$$(2.16) \quad \frac{1}{N} \text{tr} (M' \tau M \Xi)^2 = \text{tr}(\Sigma_0 T)^2 + 2 \text{tr} \Sigma_1 T \Sigma_1' T + \dots + 2 \text{tr} \Sigma_k T \Sigma_k' T + O(N^{-1})$$

To reduce Ξ^* to a matrix whose order is independent of N , let

$$(2.17) \quad \Phi(2(k+1)p \times 2(k+1)p) \\ = \begin{bmatrix} \frac{1}{\sqrt{k+1}} \Sigma_0 & \frac{1}{\sqrt{k}} \Sigma_1 & \dots & \frac{1}{\sqrt{2}} \Sigma_{k-1} & \Sigma_k \\ \frac{1}{\sqrt{k}} \Sigma_1' & \frac{1}{\sqrt{k+1}} \Sigma_0 & \dots & \frac{1}{\sqrt{3}} \Sigma_{k-2} & \frac{1}{\sqrt{2}} \Sigma_{k-1} \\ \vdots & \vdots & & \vdots & \vdots \\ \Sigma_k' & \frac{1}{\sqrt{2}} \Sigma_{k-1}' & \dots & \frac{1}{\sqrt{k}} \Sigma_1' & \frac{1}{\sqrt{k+1}} \Sigma_0 \end{bmatrix}$$

and let

$$(2.18) \quad \tau_k = (I_{k+1} \otimes T)$$

Then

$$(2.19) \quad \text{tr} \Phi \tau_k \Phi \tau_k = \text{tr}(\Sigma_O T)^2 + 2\text{tr} \Sigma_1 T \Sigma_1' T + \dots + 2\text{tr} \Sigma_k T \Sigma_k' T$$

so that

$$(2.20) \quad \phi_u(T) = \text{etr}(-\Phi \tau_k \Phi \tau_k) \{1 + o(N^{-1/2})\}$$

3. Asymptotic Approximation to $\hat{\beta}$

To find a variable whose asymptotic characteristic function is equal to that of U, let $V(2(k+1)p \times 2(k+1)p)$ have a symmetric normal distribution with (Nel (1978))

$$(3.1) \quad V = V' \sim N(O; K'_{2(k+1)p} (2\Phi \otimes \Phi) K_{2(k+1)p})$$

and partition V into $(k+1) \times (k+1)$ submatrices of order $(2p \times 2p)$.

Then

$$(3.2) \quad V_{ii} = V'_{ii} \sim N(O; K'_{2p} (2\Phi_{ii} \otimes \Phi_{ii}) K_{2p})$$

when Φ is partitioned as V. ($i=1, 2, \dots, k+1$)

Consider now the characteristic function of the sum of the diagonal matrices, V^*

$$(3.3) \quad \phi_{V^*}(T) = E \{ \text{etr} i(V_{11} + \dots + V_{k+1, k+1})T \} = \text{etr}(-\Phi \tau_k \Phi \tau_k)$$

where $T = T' (2p \times 2p)$.

This is equal to the characteristic function of U in the limit, so that

$$- \quad \frac{XX'}{N} = \Sigma_O + \frac{1}{\sqrt{N}}U \text{ will be approximated by}$$

$$(3.4) \quad \Sigma_O + \frac{1}{\sqrt{N}}V^*$$

$\hat{\beta}$ will be approximated by

$$(3.5) \quad B_n(V) = -(-\beta F + \frac{1}{\sqrt{N}} V_{12}^*) (F + \frac{1}{\sqrt{N}} V_{22}^*)^{-1}$$

$$(3.6) \quad = -(-\beta + \frac{1}{\sqrt{N}} V_{12}^* F^{-1}) (I_p - \frac{1}{\sqrt{N}} V_{22}^* F^{-1} + \frac{1}{N} (V_{22}^* F^{-1})^2 + o(N^{-3/2}))$$

so that

$$(3.7) \quad \begin{aligned} Z &= \sqrt{N}(B_n(V) - \beta) \\ &= -B + \frac{1}{\sqrt{N}} BC - \frac{1}{N} BC^2 + o(N^{-3/2}) \end{aligned}$$

where

$$(3.8) \quad B = (V_{12}^* + \beta V_{22}^*) F^{-1}$$

and

$$(3.9) \quad C = V_{22}^* F^{-1}$$

Lemma 3.1

Let $B = \sum_{i=1}^{k+1} B_{ii}$, then $B_{ii} \sim N(O; \frac{1}{k+1} \Sigma \otimes F^{-1})$ independent of B_{jj} ($j \neq i$),

and

$$(3.10) \quad B \sim N(O; \Sigma \otimes F^{-1})$$

Lemma 3.2

For B and C defined in (3.8) and (3.9),

let $H = \text{cov}(C', B)$ and $J = \text{cov}(C', C)$, then

$$(3.11) \quad H = - \sum_{s=1}^k \{ \beta^{s-1} \Sigma \otimes F^{-1} \beta^s + (F^{-1} \beta^{s-1} \Sigma \otimes \beta^s)_{(p)} \}$$

$$(3.12) \quad \begin{aligned} J &= F \otimes F^{-1} + (I_p \otimes I_p)_{(p)} + \sum_{s=1}^k \{ F \beta^{s'} \otimes \beta^s F^{-1} \\ &\quad + \beta^s F \otimes F^{-1} \beta^s + (\beta^{s'} F \beta^s F^{-1})_{(p)} + (F^{-1} \beta^s F \otimes \beta^s)_{(p)} \} \end{aligned}$$

where $(A \otimes B)_{(p)}$ denotes, for $A, B(p \times p)$, the p -row permuted matrix.

For proofs see Groenewald (1978) p. 104-109. Also Groenewald and de Waal (1978).

From (3.7) and (3.11) the expected value of $Z = \sqrt{N}(B_n(V) - \beta)$ is

$$E(Z) = \frac{1}{\sqrt{N}} E(BC) + o(N^{-3/2})$$

which yields:

$$(3.13) \quad E(Z) = -\frac{1}{\sqrt{N}} \sum_{s=1}^k \{ \Sigma \beta^{s-1'} F^{-1} (\beta^{s'} + (\text{tr} \beta^s) I_p) \} + o(N^{-3/2})$$

4. The Characteristic Function of Z

Theorem 4.1

Let $Y_t + \beta Y_{t-1} = U_t \quad t=1, 2, \dots$

where Y_t, U_t are $p \times 1$ and $U_t \sim N(0; \Sigma)$, independent of each other and let $Y_0 = 0$ and $\text{cov}(Y_t, Y_t') = F$. Then the ML estimator for $\beta(p \times p)$ is given by

$$\hat{\beta} = -\left(\sum_{t=1}^N Y_t Y_{t-1}' \right) \left(\sum_{t=1}^N Y_{t-1} Y_{t-1}' \right)^{-1}$$

Let $Z^* = \sqrt{N}(\hat{\beta} - \beta)$ with the absolute values of the characteristic roots of $-\beta$ smaller than one. Then an asymptotic approximation to the characteristic function of Z^* , is given by $\phi_Z(T)$ where $Z = \sqrt{N}(B_n(V) - \beta)$.

$$(4.1) \quad \phi_Z(T) = \text{etr}(-\frac{1}{2} \Sigma T F^{-1} T') \left\{ 1 + \frac{i}{\sqrt{N}} h_1(T) + \frac{1}{N} h_2(T) + \rho_N(T) \right\}$$

where $N^{3/2} |\rho_N(T)|$ is bounded for all T , independent of N , and where

$$(4.2) \quad h_1(T) = -\text{tr} M N T' - \sum_{s=1}^k \{ \text{tr} M \beta^{2s-1} + (\text{tr} \beta^s)(\text{tr} M \beta^{s-1}) \}$$

and

$$(4.3) \quad h_2(t) = \text{tr} M T' M T' - \frac{3}{2} (p+1) \text{tr} M T' + \frac{3}{2} \text{tr} M N^2 T' - \frac{1}{2} (h_1(T))^2 \\ + \sum_{s=1}^k \{ -2 \text{tr} M \beta^{2s} T' - 2(\text{tr} \beta^s)(\text{tr} M \beta^s T') + \text{tr} M N \beta^{2s-1} \\ + \text{tr} M \beta^{2s-1} N + (\text{tr} \beta^s)(\text{tr} M N \beta^{s-1}) + (\text{tr} \beta^s N)(\text{tr} M \beta^{s-1}) \\ + 2 \text{tr} \Sigma T \beta^{s'} F^{-1} T' M \beta^s T' - 2 \text{tr} \Sigma \beta^{s-1} F^{-1} T' M \beta^s N T' \}$$

$$\begin{aligned}
& -\frac{1}{2} \sum_{r=1}^k \sum_{s=1}^k \{ (\text{tr} M \beta^{s+r-1})^2 + 2 \text{tr} M \beta^{r-1} M \beta^{2s+r-1} \\
& + (\text{tr} \beta^{r+s}) (\text{tr} M \beta^{r-1} M \beta^{s-1}) \}
\end{aligned}$$

with

$$M = \Sigma T F^{-1}$$

$$N = -\psi F^{-1}$$

where ψ is given by (4.12), and k is chosen such that $\beta^{k+1} \rightarrow 0$, but not larger than N .

Proof:

$$\begin{aligned}
(4.4) \quad \phi_Z(T) &= E(\text{etr } i Z T') \\
&= E[\text{etr}(-i B T') \{1 + \frac{i}{\sqrt{N}} \text{tr} B C T' - \frac{1}{N} (\text{tr} B C^2 T' \\
&\quad + \frac{1}{2} (\text{tr} B C T')^2) + o(N^{-3/2})\}]
\end{aligned}$$

after expanding. To find the expectation, consider the joint distribution of B and C .

Let $G = \text{cov}(B', B)$, $Y(2p \times p) = \begin{bmatrix} B \\ C \end{bmatrix}$ and let

$y_r(2p^2 \times 1)$ be the row-packed column vector of Y . Then

$$y_r = \begin{bmatrix} b_r \\ c_r \end{bmatrix} \sim N(O; \Delta)$$

with $\Delta = \begin{bmatrix} G & H' \\ H & J \end{bmatrix}$ where H and J are given in lemma 3.2

It follows that

$$\begin{aligned}
(4.5) \quad \text{etr}(-i B T') &= \exp(-i b_r' t_r) \\
&= \exp(-i y_r' \kappa)
\end{aligned}$$

where $\kappa = \begin{bmatrix} t_r \\ 0 \end{bmatrix}$

Denoting the expansion in (4.4) by $g(y_r)$,

$$\begin{aligned} (4.6) \quad \phi_Z(T) &= E[\exp(-iy_r' \kappa) g(y_r)] \\ &= K \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} g(y_r) \exp(-\frac{1}{2} y_r' \Delta^{-1} y_r - iy_r' \kappa) dy_r \\ &= \exp(-\frac{1}{2} \kappa' \Delta \kappa) E[g(y_r)] \end{aligned}$$

where now

$$y_r \sim N(-i\Delta\kappa; \Delta)$$

with

$$(4.7) \quad E(y_r) = -i\Delta\kappa = \begin{bmatrix} -iGt_r \\ -iHt_r \end{bmatrix}$$

$$(4.8) \quad E(Y) = E \begin{bmatrix} B \\ C \end{bmatrix} = \begin{bmatrix} -iM \\ -iN \end{bmatrix}$$

where

$$(4.9) \quad M = \Sigma T F^{-1}$$

and

$$(4.10) \quad N = -\psi F^{-1}$$

where

$$(4.11) \quad \Omega = \Sigma T \beta' + \beta T' \Sigma$$

and

$$(4.12) \quad \psi = \sum_{s=1}^k \beta^{s-1} \Omega \beta'^{s-1}$$

In (4.6) we have that (Nel (1978))

$$(4.13) \quad \begin{aligned} \kappa' \Delta \kappa &= t_r' G t_r = t_r' (\Sigma \otimes F^{-1}) t_r \\ &= \text{tr} \Sigma T F^{-1} T' \end{aligned}$$

so that the characteristic function of Z can now be written as

$$\phi_Z(T) = \text{etr}(-\frac{1}{2} \Sigma T F^{-1} T') E[g(Y)]$$

where

$$B \sim N(-i \Sigma T F^{-1}, G),$$

$$C \sim N(i \psi F^{-1}, J),$$

and $\text{cov}(C', B) = H$

Now

$$(4.15) \quad g(Y) = 1 + \frac{i}{\sqrt{N}} \text{tr} B C T' - \frac{1}{N} \{i \text{tr} B C^2 T' + \frac{1}{2} (\text{tr} B C T')^2\} + o(N^{-3/2})$$

After taking expectations, the result follows.

5. The Density Function of Z

Theorem 5.1

Let Z^* be as defined in theorem 4.1, then an asymptotic approximation to the density function of Z^* is given by

$$(5.1) \quad \begin{aligned} f_Z(Z) &= (2\pi)^{-p^2/2} |\Sigma|^{-p/2} |F|^{p/2} \text{etr}(-\frac{1}{2} \Sigma^{-1} Z F Z') \{1 + \frac{1}{\sqrt{N}} g_1(Z) \\ &\quad + \frac{1}{N} g_2(Z) + o(N^{-3/2})\} \end{aligned}$$

where

$$(5.2) \quad \begin{aligned} g_1(Z) &= \sum_{s=1}^k \{(\text{tr} \beta^s) (\text{tr} \beta^{s-1} Z) + \text{tr} \beta^{2s-1} Z + 2p \text{tr} \beta^{s-1} Z F \beta' s_F^{-1} \\ &\quad - 2 \text{tr} \beta^{s-1} Z F \beta' s_Z' \Sigma^{-1} Z\} \end{aligned}$$

and

$$(5.3) \quad g_2(Z) = k_0(Z) + k_2(Z) + k_4(Z) + k_6(Z)$$

with

$$(5.4) \quad k_0(Z) = \frac{1}{2}p^2(p-1) - \sum_{s=1}^k \{p(\text{tr}\beta^{2s}) + p(\text{tr}\beta^s)^2 - 2p^2 \text{tr}F\beta'^s F^{-1}\beta^s\}$$

$$+ \sum_{s=1}^k \sum_{r=1}^k \{(2p+1)\text{tr}\Sigma\beta'^{2s+r-1}F^{-1}\beta^{r-1}$$

$$+ (\text{tr}\beta^{r+s})(\text{tr}\Sigma\beta'^{s-1}F^{-1}\beta^{r-1}) - \text{tr}\Sigma\beta'^{2s+2r-2}F^{-1}$$

$$+ 2(\text{tr}\beta^r)(\text{tr}\Sigma\beta'^{2s-1}F^{-1}\beta^{r-1}) - (\text{tr}\beta^{2s+r-1})(\text{tr}\Sigma\beta'^{r-1}F^{-1})$$

$$- 2(\text{tr}\beta^r)(\text{tr}\Sigma\beta'^{s-1}F^{-1}\beta^{r+s-1})$$

$$- 2p(p+1)\text{tr}\Sigma\beta'^{r-1}F^{-1}\beta^r F\beta'^s F^{-1}\beta^{s-1}$$

$$+ p\text{tr}\Sigma\beta'^{s-1}F^{-1}\beta^r F\beta'^s F^{-1}\beta^{r-1}$$

$$+ p(\text{tr}\beta'^s F^{-1}\beta^r F)(\text{tr}\Sigma\beta'^{s-1}F^{-1}\beta^{r-1})\}$$

$$(5.5) \quad k_2(Z) = -\frac{1}{2}(5p+1)\text{tr}\Delta - \sum_{s=1}^k \{\text{tr}\Delta\beta^{2s} + (\text{tr}\beta^s)(\text{tr}\Delta\beta^s)$$

$$+ 2p\text{tr}\Delta F\beta'^s F^{-1}\beta^s + 2p\text{tr}F^{-1}\Delta\beta^s F\beta'^s\}$$

$$+ \sum_{s=1}^k \sum_{r=1}^k \{\text{tr}Z\beta^{s-1}Z\beta^{s+2r-1}$$

$$+ \frac{1}{2}(\text{tr}\beta^{r+s})(\text{tr}Z\beta^{r-1}Z\beta^{s-1}) + \frac{1}{2}(\text{tr}\beta^{s+r-1}Z)^2$$

$$+ \frac{1}{2}(\text{tr}\beta^{2s-1}Z)(\text{tr}\beta^{2r-1}Z) + (\text{tr}\beta^r)(\text{tr}\beta^{r-1}Z)(\text{tr}\beta^{2s-1}Z)$$

$$+ \frac{1}{2}(\text{tr}\beta^r)(\text{tr}\beta^s)(\text{tr}Z\beta^{r-1})(\text{tr}Z\beta^{s-1})$$

$$+ 2p(\text{tr}\beta^s)(\text{tr}Z\beta^{s-1})(\text{tr}\beta^{r-1}ZF\beta'^r F^{-1})$$

$$- p\text{tr}\beta^{s-1}ZF\beta'^s F^{-1}\beta^{r-1}\alpha\beta'^{r-1}F^{-1} + \text{tr}F^{-1}\Delta\beta^{s-1}\Sigma\beta'^{r-1}F^{-1}\beta^s F\beta'^r$$

$$\begin{aligned}
& + \frac{1}{2}(\text{tr}\beta'^{r-1}F^{-1}\Delta\beta^{s-1}\Sigma)(\text{tr}\beta'^sF^{-1}\beta^rF) \\
& + \frac{1}{2}(\text{tr}\beta'^{s-1}F^{-1}\beta^{r-1}\Sigma)(\text{tr}\beta'^rF^{-1}\Delta\beta^sF) + \text{tr}Z\beta^{2s+r-2}a\beta'^{r-1}F^{-1} \\
& + 2p\text{tr}\beta^{s+r-1}a\beta'^{r-1}Z'\beta'^{s-1}F^{-1} \\
& + 2p(\text{tr}Z\beta^{2s-1})(\text{tr}\beta'^{r-1}ZF\beta'^rF^{-1}) - 4\text{tr}\beta'^{2s+r-1}F^{-1}\Delta\beta^{r-1}\Sigma \\
& + 2p^2(\text{tr}\beta'^{r-1}ZF\beta'^rF^{-1})(\text{tr}\beta'^{s-1}ZF\beta'^sF^{-1}) \\
& - \text{tr}\beta^{s+r-1}a\beta'^{r-1}F^{-1}\beta^{s-1}Z \\
& + 4p\text{tr}\beta'^sF^{-1}\Delta\beta^{s-1}\Sigma\beta'^{r-1}F^{-1}\beta^rF \\
& - 4(\text{tr}\beta^r)(\text{tr}\beta'^{s-1}F^{-1}\Delta\beta^{s+r-1}\Sigma) \}
\end{aligned}$$

$$\begin{aligned}
(5.6) \quad k_4(Z) = & \text{tr}\Delta^2 + 2 \sum_{s=1}^k \text{tr}F^{-1}\Delta\beta^s\Delta F\beta'^s \\
& - \sum_{s=1}^k \sum_{r=1}^k \{ 2\text{tr}F^{-1}\Delta\beta^{s-1}Z\beta'^{r-1}a\beta'^{s+r-1} \\
& + \frac{1}{2}\text{tr}F^{-1}\Delta\beta^{s-1}a\beta'^{s-1}F^{-1}\beta^{r-1}a\beta'^{r-1} \\
& + 2\text{tr}F^{-1}\Delta\beta^rF\beta'^sF^{-1}\Delta\beta^{s-1}\Sigma\beta'^{r-1} \\
& + 2(\text{tr}Z\beta^{2s-1})(\text{tr}F^{-1}\Delta\beta^{r-1}ZF\beta'^r) \\
& + 2(\text{tr}\beta^s)(\text{tr}Z\beta^{s-1})(\text{tr}F^{-1}\Delta\beta^{r-1}ZF\beta'^r) \\
& + 4p(\text{tr}\beta^{s-1}ZF\beta'^sF^{-1})(\text{tr}F^{-1}\Delta\beta^rFZ'\beta'^{r-1}) \}
\end{aligned}$$

$$(5.7) \quad k_6(Z) = 2 \sum_{s=1}^k \sum_{r=1}^k (\text{tr}F^{-1}\Delta\beta^rFZ'\beta'^{r-1})(\text{tr}F^{-1}\Delta\beta^sFZ'\beta'^{s-1})$$

$$(5.8) \quad \alpha = ZF\beta' + \beta FZ'$$

$$(5.9) \quad \Delta = FZ'\Sigma^{-1}Z$$

Proof:

Applying the inverse Fourier transformation to the characteristic function of

$Z = \sqrt{N} (B_N(V) - \beta)$, we have

$$f_Z(Z) = (2\pi)^{-p^2} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \text{etr}(-iZT') \phi_Z(T) dT$$

Substituting (4.1) and after some simplification it follows that

$$(5.10) \quad f_Z(Z) = (2\pi)^{-p^2/2} |\Sigma|^{-p/2} |F|^{p/2} \text{etr}(-\frac{1}{2} \Sigma^{-1} Z F Z') E \left\{ 1 + \frac{i}{\sqrt{N}} h_1(T) + \frac{1}{N} h_2(T) + o(N^{-3/2}) \right\}$$

where

$$T \sim N(-i\Sigma^{-1}ZF; \Sigma^{-1} \otimes F)$$

The result follows after taking expected values (Groenewald (1978) p. 120-125).

In the univariate case, we have $\text{var}(U_t) = \sigma$, $\text{var}(Y_t) = f$, where

$$(5.11) \quad \sigma = f(1-\beta^2)$$

Then

$$(5.12) \quad g_1(Z) = \frac{2\beta}{1-\beta^2} \left(2Z - \frac{Z^3}{1-\beta^2} \right)$$

$$(5.13) \quad g_2(Z) = \frac{3}{1-\beta^2} \left(\frac{4\beta^2}{1-\beta^2} - 1 \right) Z - \frac{1}{(1-\beta^2)^2} \left(\frac{14\beta^2}{1-\beta^2} - 1 \right) Z^4 + \frac{2\beta^2}{(1-\beta^2)^2} Z^6$$

with

$$(5.14) \quad E(Z) = -\frac{2}{\sqrt{N}}\beta + o(N^{-3/2})$$

$$(5.15) \quad \text{var}(Z) = (1-\beta^2) + \frac{3}{N}(3+13\beta^2) + o(N^{-2})$$

REFERENCES

- ANDERSON, T.W. (1958). An Introduction to Multivariate Statistical Analysis. *New York, Wiley, 1958.*
- ANDERSON, T.W. (1959). On Asymptotic Distributions of Estimates of Parameters of Stochastic Difference Equations. *Ann. Mat. Statist.* **30**, 676-687.
- ANDERSON, T.W. (1971). The Statistical Analysis of Time Series. *New York, Wiley, 1971.*
- ANDERSON, T.W. & RUBIN, H. (1950). The Asymptotic Properties of Estimates of the Parameters of a single Equation in a Complete System of Stochastic Equations. *Ann. Math. Statist.* **21**, 570-582.
- GROENEWALD, P.C.N. (1978). On Asymptotic Distributions of certain Regression Matrices with an Application to Multiple Time Series Analysis. *Unpublished Ph.D. thesis. Dept. of Math. Stat., University of the OFS.*
- GROENEWALD, P.C.N. and DE WAAL, D.J. (1978). An asymptotic distribution for the coefficient in a multiple time series. *Tech. report no. 31, Dept. Math. Statistics, UOFS.*
- NEL, D.G. (1978). On the symmetric multivariate normal distribution and the asymptotic expansion of a Wishart matrix. *South African Statist. J.* **12**, 145-159.

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